

Lab-9 Submission

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Task 1: Discount Calculator

Question: Write a Python program to calculate the price after applying discount.

Code :

```
# task1_calculate_discount.py
```

```
def calculate_discount_original(price, discount_rate):
```

```
    # Original function (no validation)
```

```
    return price - (price * discount_rate / 100)
```

```
def calculate_discount_ai(price, discount_rate): #
```

```
Auto-generated style comments (example) #
```

```
Validate input values to avoid wrong calculation
```

```
    if price < 0:
```

```
        raise ValueError("price must be >= 0")
```

```
    if not (0 <= discount_rate <= 100):
```

```
        raise ValueError("discount_rate must be between 0 and 100")
```

```
    # Calculate discount amount and subtract from price
```

```
    discount_amount = price * discount_rate / 100
```

```
    return price - discount_amount
```

```

def calculate_discount_final(price, discount_rate):
    """Calculate the price after applying a percentage discount.

    Args:
        price (float): Original price (must be >= 0).
        discount_rate (float): Discount percent (0 - 100).

    Returns:
        float: Price after discount applied.

    Raises:
        ValueError: If inputs are invalid.
    """
    # Input validation (avoid silent errors)
    if price < 0:
        raise ValueError("price must be >= 0")
    if not (0 <= discount_rate <= 100):
        raise ValueError("discount_rate must be between 0 and 100")

    # Compute discount amount and return final price
    discount_amount = price * discount_rate / 100
    return price - discount_amount

print(calculate_discount_final(200, 10))

if __name__ == "__main__":
    price = float(input("Enter price: "))    discount =
    float(input("Enter discount rate (%): "))    final_price =
    calculate_discount_final(price, discount)    print("Final
    price after discount:", final_price)

```

Task 2: Library Management System

Question: Create functions to add books and issue books in a library.

Code :

```
def add_book(title, author, year):
```

```
    """Add a new book record to the library.
```

Args:

title (str): Book title.

author (str): Author name.

year (int): Year of publication.

Returns:

dict: Book details with id.

```
    """ book_id = 1 return {"book_id": book_id, "title": title, "author":  
author, "year": year}
```

```
def issue_book(book_id, user_id):
```

```
    """Issue a book to a user.
```

Args:

book_id (int): ID of the book.

user_id (int): ID of the user.

Returns:

bool: True if issued successfully.

```
    """
```

```
return True if
```

```
__name__ ==
```

```
"__main__":
```

```

book =
add_book("P
ython
Basics",
"Guido",
2020)
print("Book
Added:",
book)
issued =
issue_book(b
ook["book_id
"], 101)
print("Book
Issued:",
issued)

```

Output:

```

Book Added: {'book_id': 1, 'title': 'Python Basics', 'author': 'Guido',
'year': 2020}
Book Issued: True

```

Task 3: Student Grade System

Question: Write a Python function to calculate grade based on marks.

Code :

```

def calculate_grade(marks):
    """Calculate grade based on marks.

```

Args:

marks (int): Marks scored by student.

Returns:

str: Grade (A, B, C, D, F).

"""

if marks >= 90:

return "A" elif

marks >= 75:

return "B"

elif marks >= 60:

return "C"

elif marks >= 40:

return "D" else:

return "F"

if __name__ == "__main__":

print("Grade for 95:", calculate_grade(95))

print("Grade for 72:", calculate_grade(72))

print("Grade for 50:", calculate_grade(50))

print("Grade for 30:", calculate_grade(30))

Output:

Grade for 95: A

Grade for 72: B Grade

for 50: C

Grade for 30: F

Task 4: Student Management System

Question: Write a Python program to manage student records (add, display, search)

Code :

students = [] # list to store student records

```

def add_student(roll_no, name, marks): """Add new student
to the system""" student = {"roll_no": roll_no, "name":
name, "marks": marks} students.append(student)

def display_students():
    """Display all students""" if
not students:

        print("No students found.")
    else:        for s in
students:

        print(f"Roll No: {s['roll_no']}, Name: {s['name']}, Marks: {s['marks']}")

def search_student(roll_no):
    """Search student by roll number"""

    for s in students:        if
s["roll_no"] == roll_no:

        return s

    return None

if __name__ == "__main__":
    # sample data
    add_student(1, "Ravi", 85)
    add_student(2, "Priya", 92)
    add_student(3, "Arjun", 60)

    print("\nAll Students:")
    display_students()

```

```

print("\nSearching Roll No 2:")

result = search_student(2)

if result:

    print(f"Found: {result}")

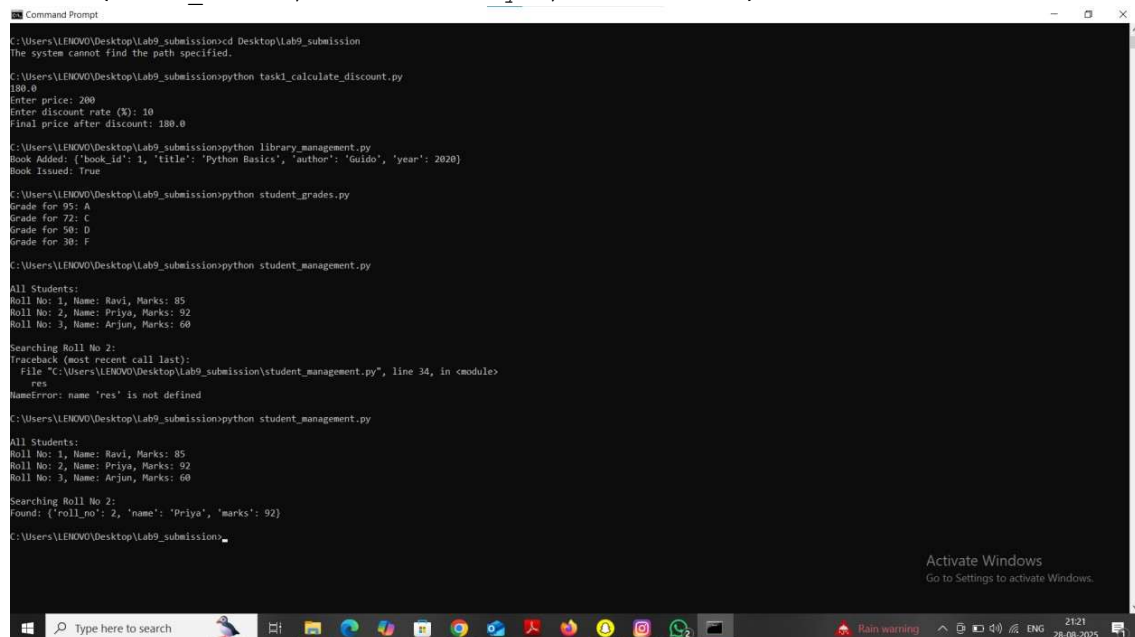
else:
    print("Student not
found")

```

Output:

All Students:
Roll No: 1, Name: Ravi, Marks: 85 Roll
No: 2, Name: Priya, Marks: 92 Roll No:
3, Name: Arjun, Marks: 60

Searching Roll No 2:
Found: {'roll_no': 2, 'name': 'Priya', 'marks': 92}



```

C:\Users\LENOVO\Desktop\Lab9_submission>cd Desktop\Lab9_submission
The system cannot find the path specified.

C:\Users\LENOVO\Desktop\Lab9_submission>python task1_calculate_discount.py
180.0
Enter price: 200
Enter discount rate (%): 10
Final price after discount: 180.0

C:\Users\LENOVO\Desktop\Lab9_submission>python library_management.py
Book Added: {'book_id': 1, 'title': 'Python Basics', 'author': 'Guido', 'year': 2020}
Book Issued: True

C:\Users\LENOVO\Desktop\Lab9_submission>python student_grades.py
Grade for 95: A
Grade for 72: C
Grade for 50: D
Grade for 30: F

C:\Users\LENOVO\Desktop\Lab9_submission>python student_management.py
All Students:
Roll No: 1, Name: Ravi, Marks: 85
Roll No: 2, Name: Priya, Marks: 92
Roll No: 3, Name: Arjun, Marks: 60

Searching Roll No 2:
Traceback (most recent call last):
  File "C:\Users\LENOVO\Desktop\Lab9_submission\student_management.py", line 34, in <module>
    res
NameError: name 'res' is not defined

C:\Users\LENOVO\Desktop\Lab9_submission>python student_management.py
All Students:
Roll No: 1, Name: Ravi, Marks: 85
Roll No: 2, Name: Priya, Marks: 92
Roll No: 3, Name: Arjun, Marks: 60

Searching Roll No 2:
Found: {'roll_no': 2, 'name': 'Priya', 'marks': 92}

C:\Users\LENOVO\Desktop\Lab9_submission>

```

